

Least-Squares Regression

Agenda

- + Today:
 - + How do we estimate coefficients in a linear regression model?
 - + More on prediction
- + Later this week
 - + Prediction and correlation
 - + Activity: practicing statistical communication

Recap: regression

Regression: model the relationship between predictor x and response y

Regression model: $y = f(x) + \varepsilon$

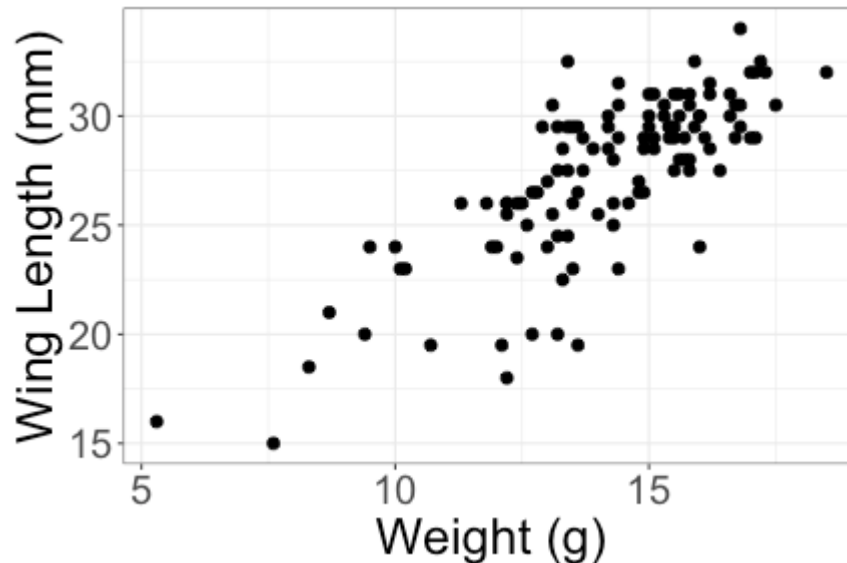
+ f = systematic information that x provides about y

+ ε = random noise

Linear regression: $y = \beta_0 + \beta_1 x + \varepsilon$

Recap: linear regression

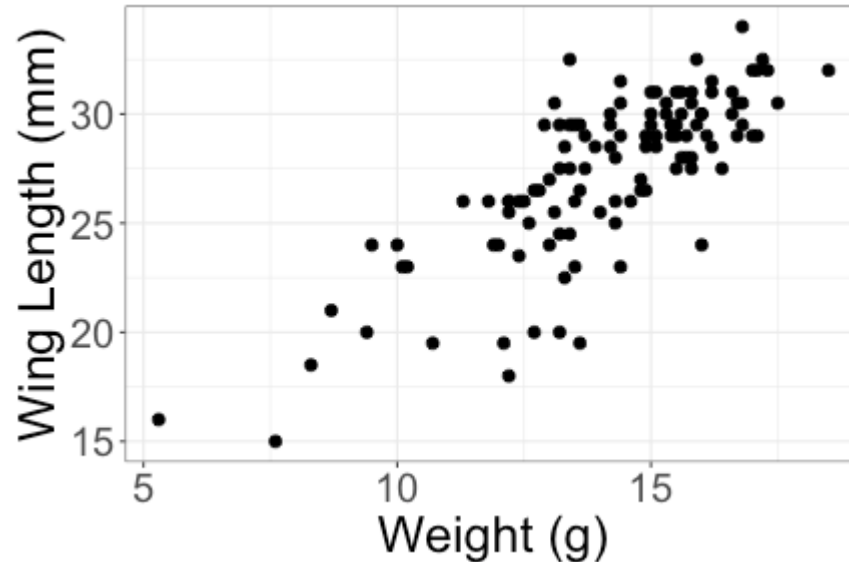
Observed data (sample): Weight and wing length for 116 sparrows



Regression model (population):

$$\text{WingLength} = \beta_0 + \beta_1 \text{Weight} + \varepsilon$$

Recap: linear regression



Estimated regression line (from sample):

$$\widehat{\text{WingLength}} = 8.76 + 1.31 \text{ Weight}$$

+ $\hat{\beta}_0 = 8.76$

+ $\hat{\beta}_1 = 1.31$

↑
estimated wing length

Recap: interpreting regression coefficients

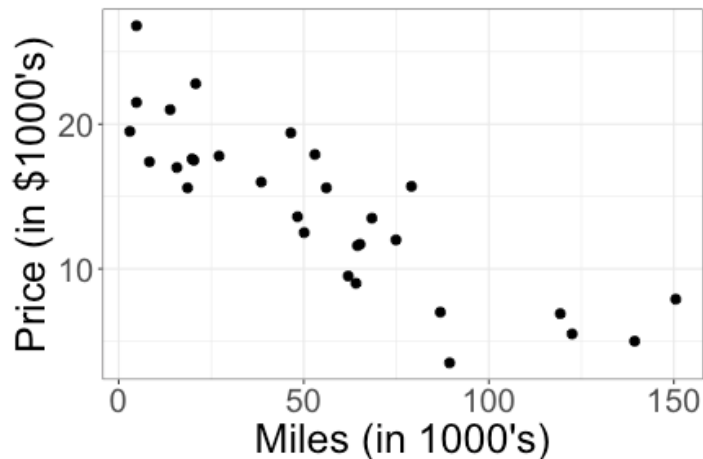
$$\widehat{\text{WingLength}} = 8.76 + 1.31 \text{ Weight}$$

Interpreting $\hat{\beta}_0$: If weight = 0g, we predict a wing length of 8.76mm.

Interpreting $\hat{\beta}_1$: An increase of 1g in weight is associated with an increase of 1.31mm, on average, in wing length

Concept check

Interested in the relationship between mileage and price for used Honda Accords. We will use linear regression.



Write the population regression model:

(A) $\text{price} = \beta_0 + \beta_1 \text{miles}$

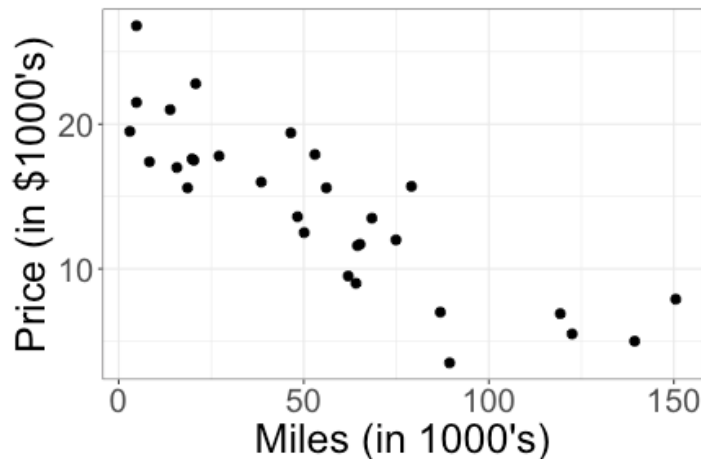
(B) $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

(C) $\widehat{\text{price}} = \hat{\beta}_0 + \hat{\beta}_1 \text{miles}$

(D) $\widehat{\text{price}} = \hat{\beta}_0 + \hat{\beta}_1 \text{miles} + \varepsilon$

Concept check

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Write the population regression model:

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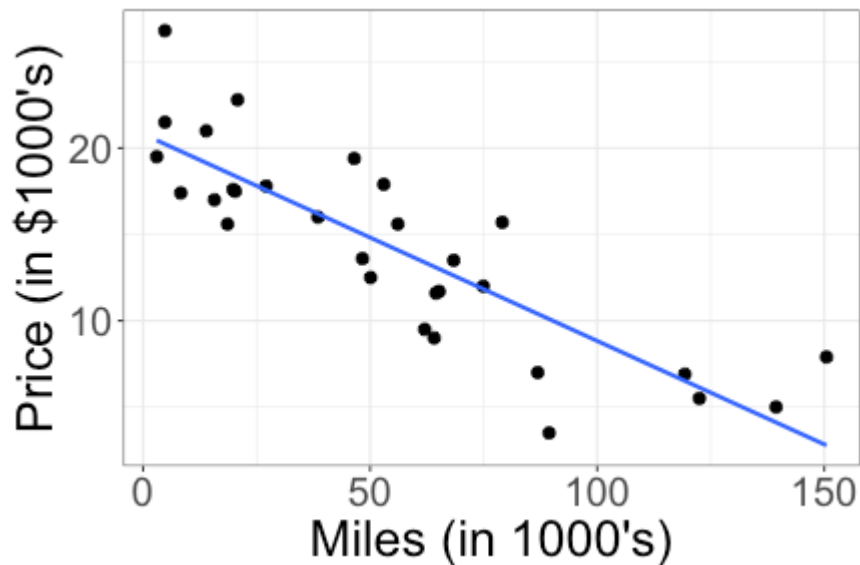
(B) $\text{price} = \beta_0 + \beta_1 \text{miles} + \varepsilon$

(C) $\widehat{\text{price}} = \hat{\beta}_0 + \hat{\beta}_1 \text{miles}$

(D) $\widehat{\text{price}} = \hat{\beta}_0 + \hat{\beta}_1 \text{miles} + \varepsilon$

Answer: (B)

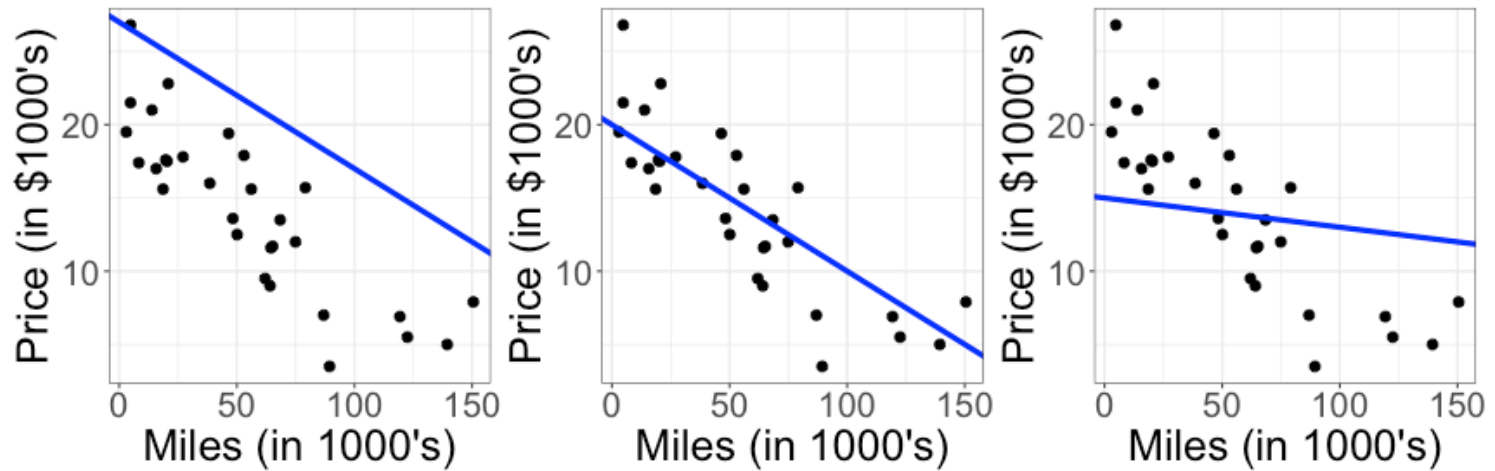
Linear regression



Estimated line (from sample): $\widehat{\text{price}} = 20.81 - 0.12 \text{ miles}$

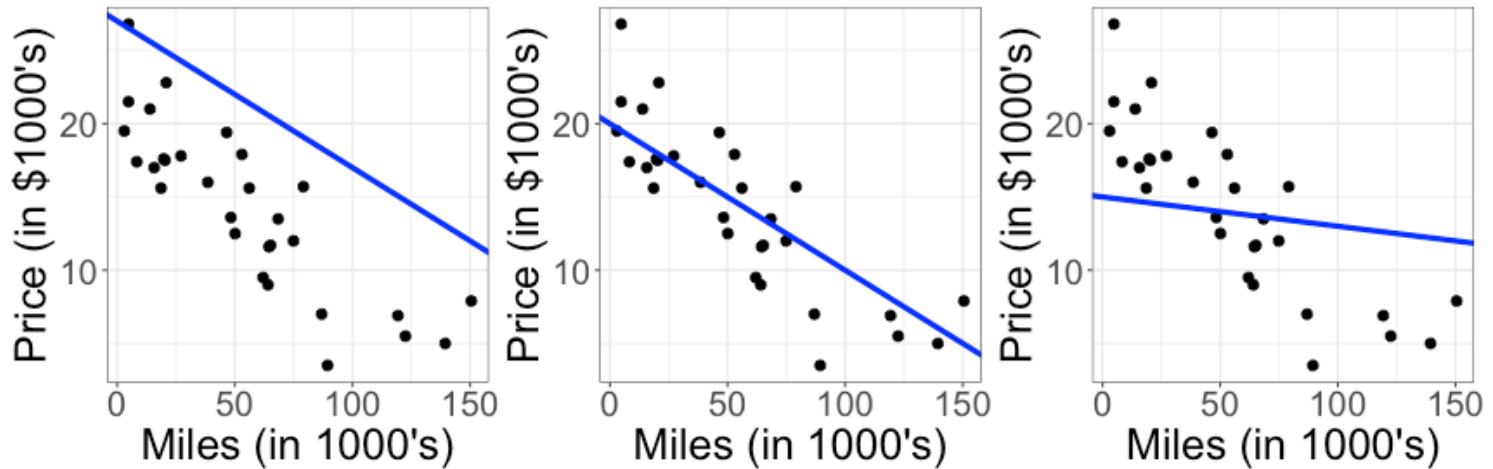
How do we calculate $\hat{\beta}_0$ and $\hat{\beta}_1$?

Choosing between different lines



Which one looks like the best choice?

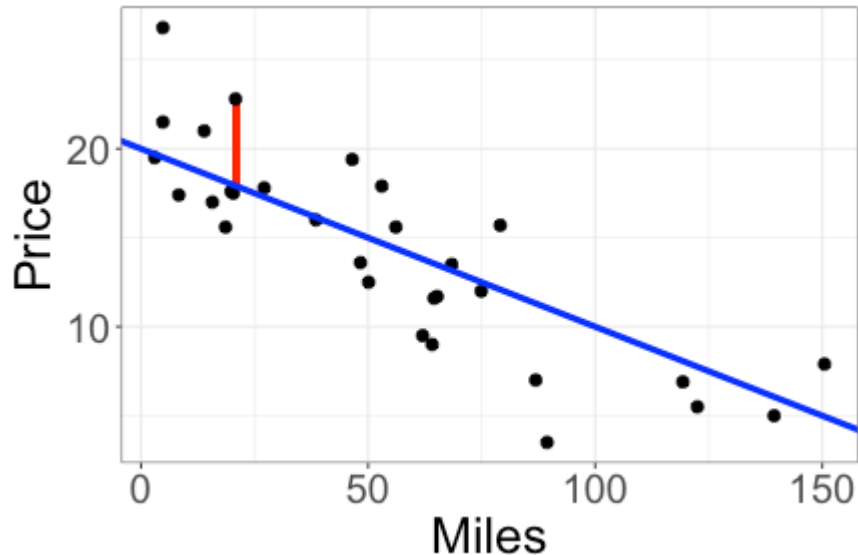
Choosing between different lines



Which one looks like the best choice?

Intuitively, want the line to "go through" the points. Can we formalize this?

Residuals



Car with mileage = 20.8

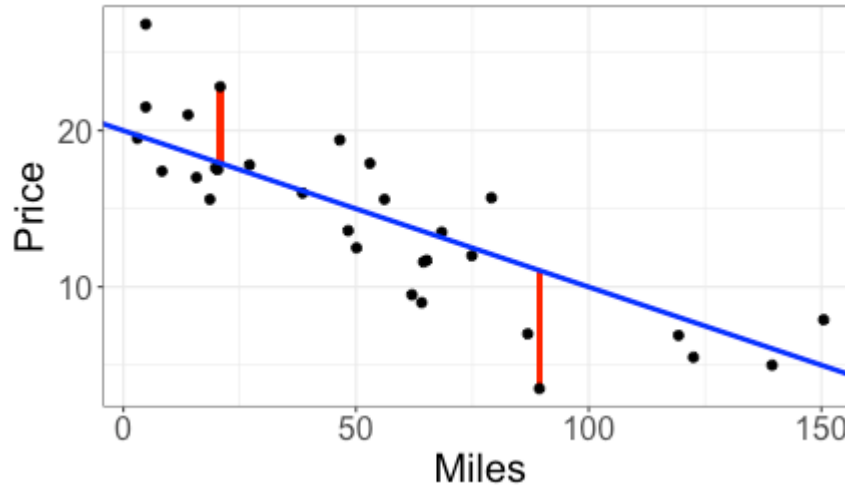
+ Prediction: $\widehat{Price} = 18$

+ Observed: $Price = 22.8$

Residual: $Price - \widehat{Price} = 22.8 - 18 = 4.8$

idea: want residuals generally close to 0

Residuals



+ Prediction: $\widehat{Price} = 18$

+ Observed: $Price = 22.8$

$$Price - \widehat{Price} = 4.8$$

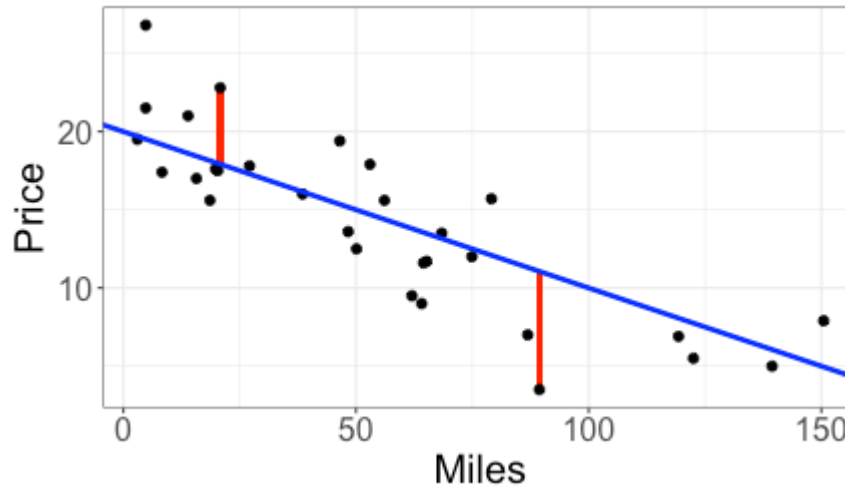
+ Prediction: $\widehat{Price} = 11.06$

+ Observed: $Price = 3.5$

$$Price - \widehat{Price} = -7.56$$

Which of these predictions is closer?

Residuals



+ Prediction: $\widehat{Price} = 18$

+ Observed: $Price = 22.8$

$$Price - \widehat{Price} = 4.8$$

+ Prediction: $\widehat{Price} = 11.06$

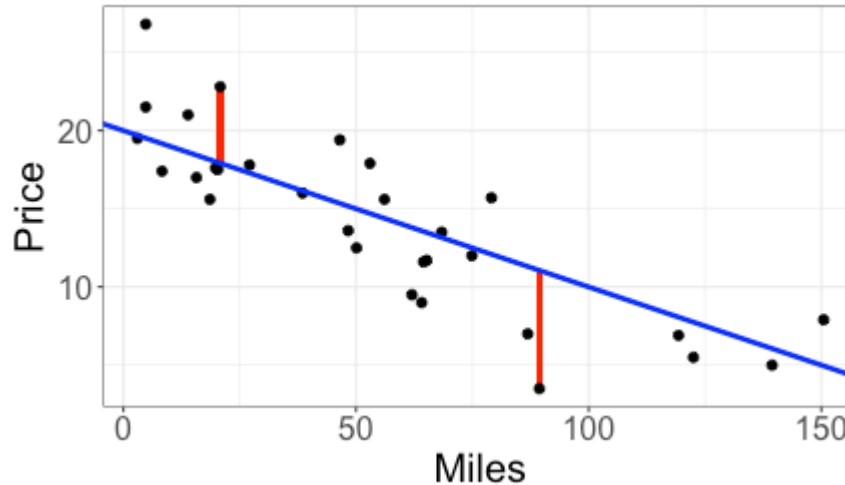
+ Observed: $Price = 3.5$

$$Price - \widehat{Price} = -7.56$$

What does a *positive* residual mean?

underestimate

Residuals



+ Prediction: $\widehat{Price} = 18$

+ Observed: $Price = 22.8$

$$Price - \widehat{Price} = 4.8$$

+ Prediction: $\widehat{Price} = 11.06$

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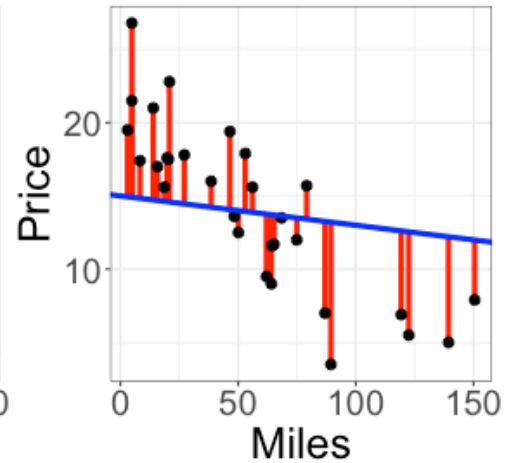
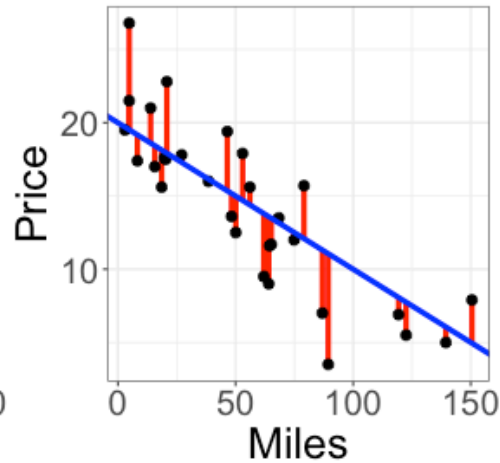
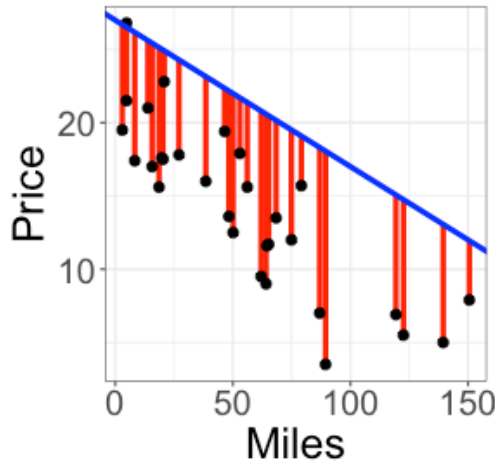
$$Price - \widehat{Price} = -7.56$$

What does a *negative* residual mean?

over estimated

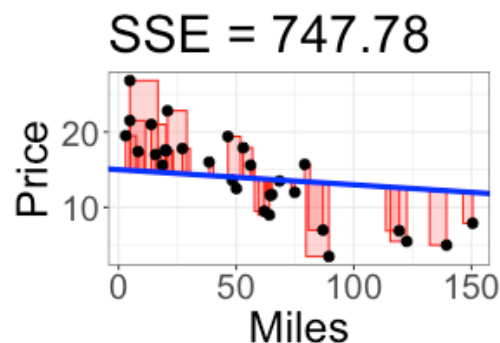
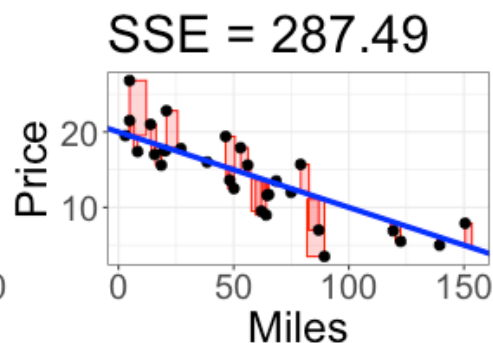
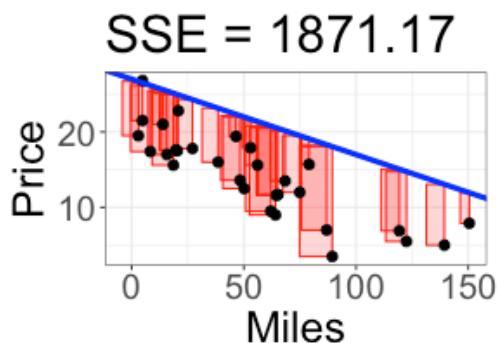
Residuals

- residuals are generally close to 0
- points are scattered above & below line



Which of these lines looks like the best fit? Why?

Sum of the squared residuals



Sum of squared residuals (SSE) = $\sum_{i=1}^n (y_i - \hat{y}_i)^2$

(aka Sum of squared errors)

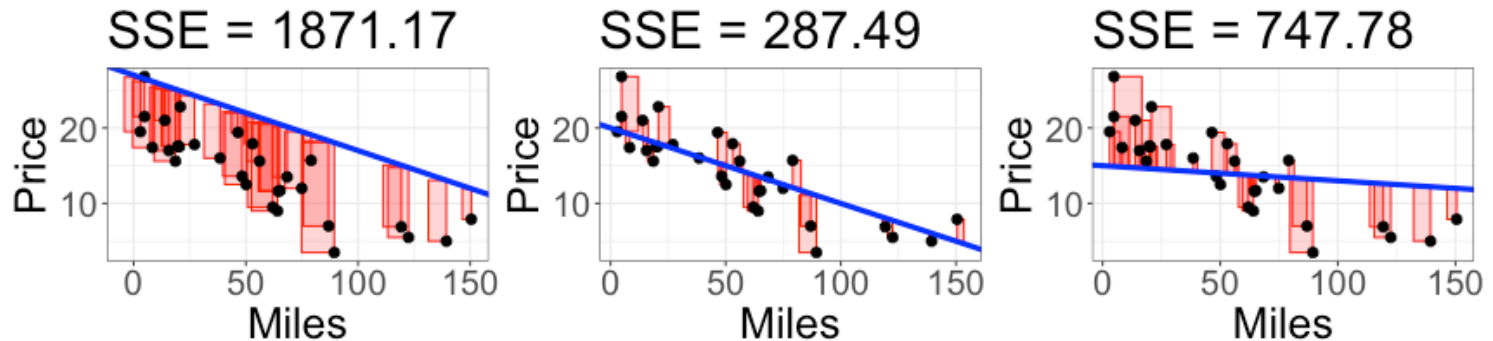
(aka residual sum of squares, RSS)

add everything up

residual

squared residual

Least Squares



Sum of squared residuals (SSE) =

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2 = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize SSE.

$$\begin{aligned}\hat{y} &= \hat{\beta}_0 + \hat{\beta}_1 x \\ y - \hat{y} &= y - (\hat{\beta}_0 + \hat{\beta}_1 x) = y - \hat{\beta}_0 - \hat{\beta}_1 x\end{aligned}$$

Class activity, Part I

Spend a couple minutes trying out different potential values for the estimated slope and intercept.

https://sta112-f25.github.io/class_activities/ca_06.html

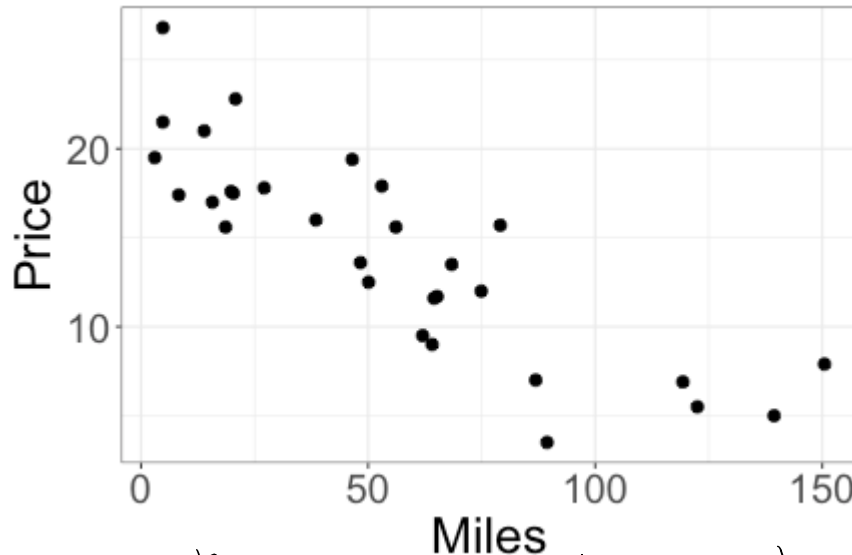
9.6	, 1.25	630
6.1	1.5	647
6.108	1.494	647

(from R)

$\hat{\beta}_0 = 8.76$ $\hat{\beta}_1 = 1.31$ 627.7

Downsides: can't test every value

Fitting linear regression in R



linear model

↓ separate response { explanatory variable

```
lm(Price ~ Mileage, data = AccordPrice)
```

- + `lm`: function used for regression ("linear model")
- + `Price`: response variable
- + `Mileage`: predictor

Fitting linear regression in R

```
lm(Price ~ Mileage, data = AccordPrice)
```

```
##  
## Call:  
## lm(formula = Price ~ Mileage, data = AccordPrice)  
##  
## Coefficients:  
## (Intercept)      Mileage  
##      20.8096      -0.1198
```

+ $\hat{\beta}_0 = 20.81$

+ $\hat{\beta}_1 = -0.12$

Class activity, Part II

https://sta112-f25.github.io/class_activities/ca_06.html

Q4

$$8.755 + 1.313 * 10$$

Calculating $\hat{\beta}_0$ and $\hat{\beta}_1$

$$SSE = \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize SSE

- + Option 1: search for $\hat{\beta}_0, \hat{\beta}_1$ like in class activity Part I

Calculating $\hat{\beta}_0$ and $\hat{\beta}_1$

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- + not efficient
 - + not what R does

Calculating $\hat{\beta}_0$ and $\hat{\beta}_1$

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Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize SSE

- + Option 1: search for $\hat{\beta}_0, \hat{\beta}_1$ like in class activity Part I
- + not efficient
 - + not what R does
- + Option 2: closed-form equation?

Equations for $\hat{\beta}_0$ and $\hat{\beta}_1$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

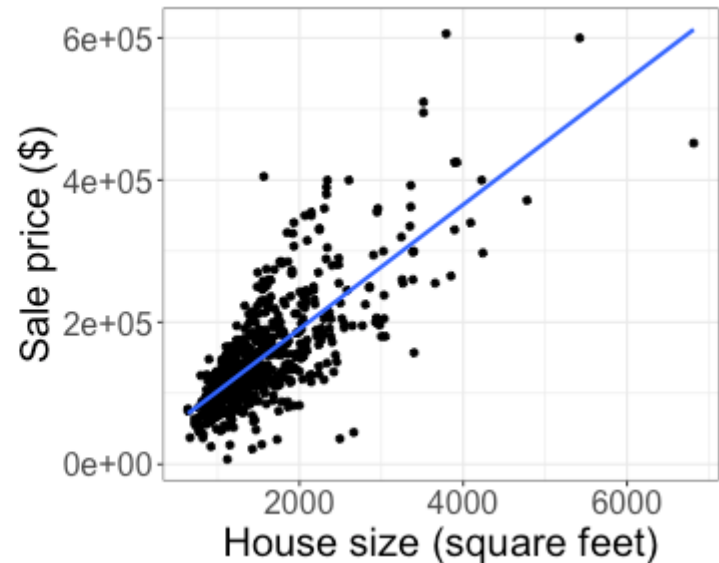
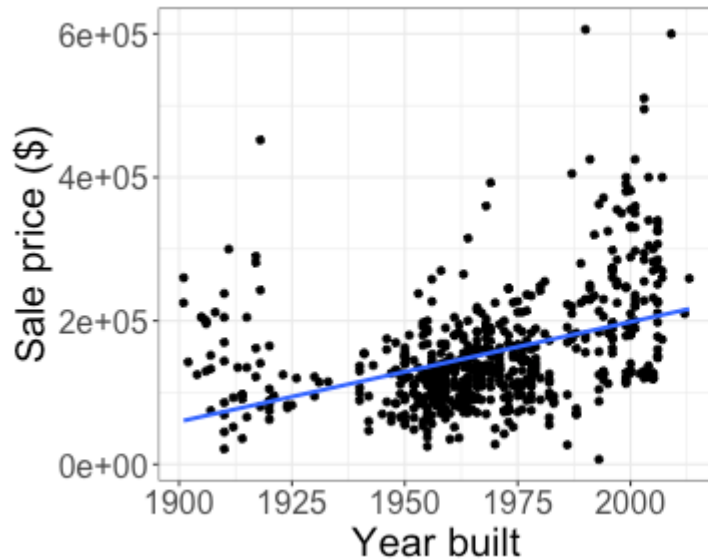
where

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Making predictions

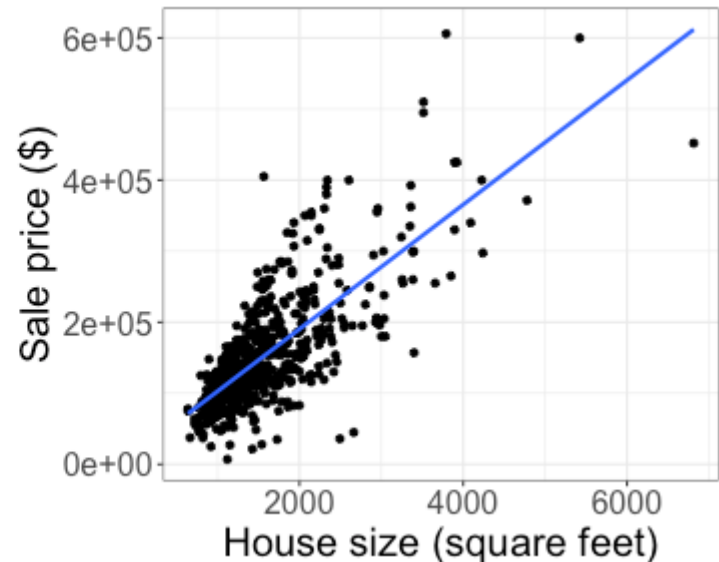
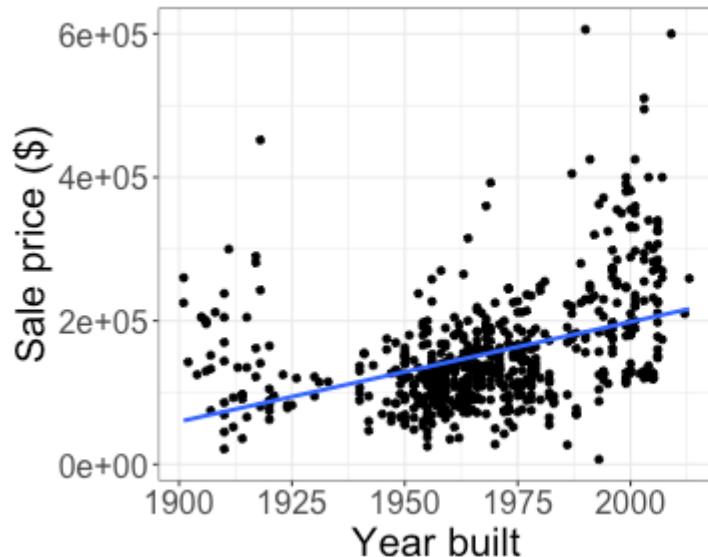
Goal: predict a house's sale price using either its year built or its size



Which of the two variables would you prefer to use?

Making predictions

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



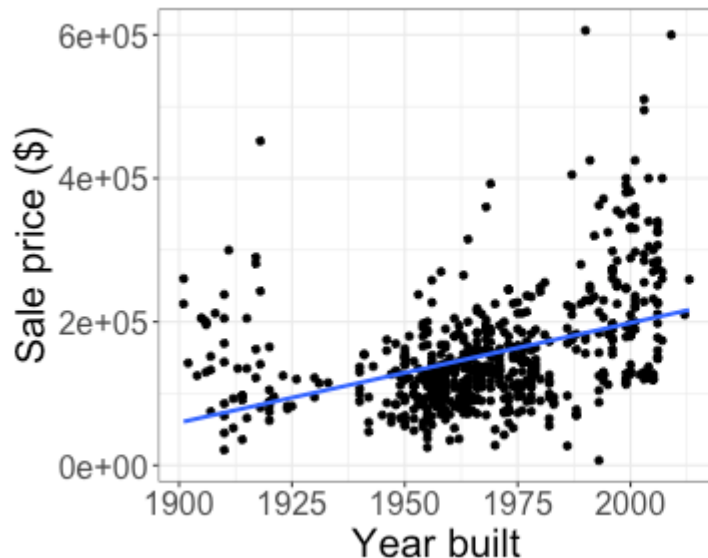
SSE: 3.41×10^{12}

SSE: 1.74×10^{12}

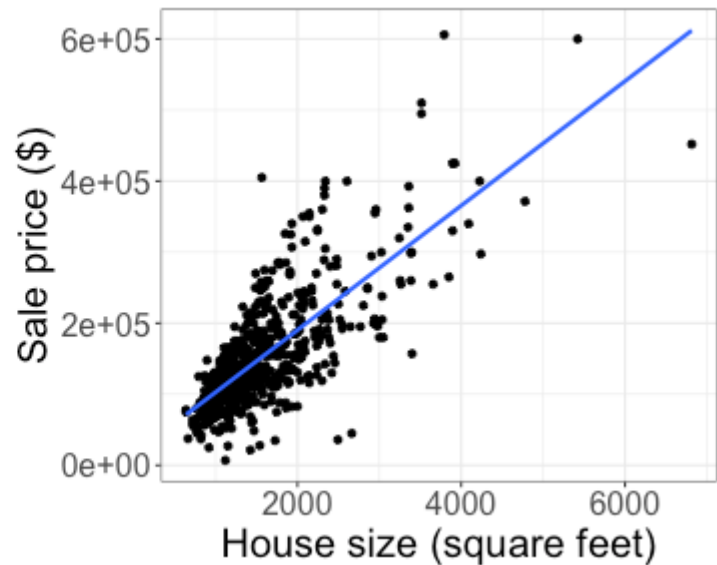
SSE measures how close, overall, our predictions are to the observed response!

Making predictions

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$



SSE: 3.41×10^{12}



SSE: 1.74×10^{12}

Why are these numbers so *large*?

Making predictions

$$SSE = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

- + Depends on the scale of the response variable
- + Depends on the *number* of observations
- Depends on the strength of the relationship